# Exact-Quota Mixed-Precision FIR Coefficient Quantization: Dynamic Programming, FPGA Validation, and Feasibility Analysis of Response-Aware Allocation

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***Abstract*—This technical report documents the design, analysis, and experimental evaluation of exact-quota mixed-precision coefficient quantization for FIR filters. The central problem assigns each of $N$ fixed coefficients one of three binary widths—16-bit, 8-bit, or 4-bit—subject to exact count constraints, and minimizes the total squared coefficient quantization error. This is equivalent to a unit-capacity bipartite transportation problem, a fact that limits the algorithmic novelty claim: the dynamic programming (DP) solution presented here is a structure-specific realization of a known cost-constrained DP class, not a new optimization principle.**

**The report documents validation across three levels: (1) software—29 exhaustive consistency checks, 108 test configurations comparing DP, greedy, and MILP baselines; (2) FPGA—6 synthesis/route experiments for an 8-tap filter on Xilinx Artix-7, and 4 synthesis/route experiments for a 48-tap symmetric-folded filter; (3) physical hardware—942 bit-exact functional comparisons across three configurations on an ACX750 development board. A 48-tap robustness study covers 15 design/quota combinations from 5 filter specifications; DP meets a $-40$ dB stopband limit in 2 of 15 cases, greedy in 0. A response-aware allocation branch was explored; a 48-tap binary-product MILP reformulation was manually stopped after about 93 seconds because its root relaxation showed negligible progress toward proving optimality. The run returned no solver exit flag or final allocation, and the branch was closed under the predeclared slow-progress stopping condition. All negative results are reported. No novelty claim is made.**

***Index Terms*—FIR filter, coefficient quantization, dynamic programming, mixed-precision, FPGA, assignment problem.**

## I. Introduction

Finite-word-length implementation of FIR filters requires assigning a coefficient precision to each tap. When the available hardware supports a discrete set of widths—for example, 16-bit, 8-bit, and 4-bit multipliers—and the system specifies exact counts at each width, the assignment becomes a combinatorial optimization problem. The natural objective in coefficient-domain design is to minimize the total squared coefficient quantization error.

This report studies the exact-quota version of that problem: given $N$ tap coefficients and integer counts $n_{16}$, $n_8$, $n_4$ with $n_{16} + n_8 + n_4 = N$, assign each coefficient a width from $\{16, 8, 4\}$ to minimize

$$\min_{\mathbf{w}} \sum_{i=1}^{N} \big(h_i - Q(h_i, w_i)\big)^2 \tag{1}$$

subject to $|\{i : w_i = w\}| = n_w$ for each width $w$.

A DP recurrence processes coefficients in order and achieves the global minimum for (1). The formulation is equivalent to a unit-capacity bipartite transportation problem or binary assignment MILP, a relationship established in Section III and verified empirically in Section V.

The report is organized as follows. Section II reviews related work including the closest prior formulations. Section III states the problem and its equivalences. Section IV describes the DP and greedy algorithm. Section V reports software validation. Section VI describes the FPGA case studies. Section VII reports the robustness study. Section VIII documents the response-aware allocation investigation and its negative result. Section IX concludes.

## II. Related Work

Several lines of prior work are directly relevant. Twelve papers were reviewed in full in the project literature assessment, and six additional works received abstract-level screening. The reviewed set is targeted, not an exhaustive search of the field.

### A. Finite-Word-Length FIR Design

Kodek and Steiglitz [1] analyzed length-wordlength tradeoffs for FIR filters, proving a nonzero lower bound on weighted Chebyshev error for fixed coefficient wordlength. Their complexity proxy $Nb$ motivates joint optimization; their analysis assumes a common wordlength, not per-coefficient quotas.

Lim, Parker, and Constantinides [2] minimized FIR designs over discrete coefficient-value spaces using integer linear and quadratic programming, comparing discrete-space designs with rounded infinite-wordlength designs. Lim [3] extended this to normalized peak-ripple minimization, using MILP methods with passband gain as a design variable.

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### B. SPT-Based and Hardware-Aware Coefficient Allocation

Several methods allocate signed-power-of-two (SPT) terms across coefficients. Llorens, Hadjicostis, and Ni [4] formulated FIR coefficient quantization under a scalar total integer-cost constraint using a constrained-shortest-path DP. Their alpha=2 case relates coefficient-vector error to frequency-response MSE via Parseval's theorem. For exact three-width quotas, a radix encoding ($r_{16} = B + 1$, $r_8 = 2$, $r_4 = 1$, $B = N + 1$) reduces the quota DP to a scalar exact-cost DP, so the present quota DP is a specialized realization of Llorens et al.'s framework, not a new optimization class.

Chen and Willson [5] used a trellis (Viterbi-like) search for SPT term allocation against frequency-response criteria; their survivor pruning does not guarantee global optimality. Li, Lim, Lian, and Song [6] allocated a global SPT-term budget with variable per-coefficient counts using greedy sequential allocation; the authors describe the result as suboptimal.

Aktan, Yurdakul, and Dündar [7] used branch-and-bound (FIRGAM) with LP for coefficient bounds to reduce SPT count while meeting frequency specifications; worst-case runtime is exponential.

Kuo, Lin, and Liu [8] integrated successive SPT allocation with common-subexpression elimination under an exact global post-CSE addition budget, reporting 0.35-$\mu$m CMOS synthesis results.

### C. Subexpression-Space and Structure Optimization

Yu and Lim [9] optimized FIR coefficients directly in a discrete subexpression space using branch-and-bound MILP, minimizing frequency-response ripple subject to an adder budget. This is a direct coefficient-space search, not a two-stage quantize-then-CSE pipeline.

Blad and Gustafsson [10] used ILP to optimize partial-product reduction trees and pipeline registers for fixed coefficient representations, targeting 90-nm CMOS. Their decision variables are hardware structural elements, not coefficient widths.

### D. Mixed-Precision FIR Allocation

Fang, Chen, Chen, Wang, and You [11] proposed a PSO-based bit-allocation framework applied to FIR filter design with a frequency-domain minimax objective and aggregate bit-consumption constraint. Their constraint is an inequality on total bits ($\sum b_n \le N\bar{b}$), not exact per-width counts. Their allocation assumes symmetric widths (linear-phase structure). This is the closest prior mixed-precision FIR allocation paper; it differs from the present work in constraint type, objective, and solution method.

Sarma, Sai, and Chatterjee [12] used a coefficient-magnitude threshold to select exact or LUT-approximate multipliers under an output-error budget on a Zynq UltraScale+ radar DDC implementation. Their quantization model (approximate arithmetic) differs from fixed-point coefficient widths.

### E. Neural-Network Mixed-Precision

Yang and Jin's FracBits [13] assigns fractional bit-widths in neural-network quantization. It is included as methodological context; it is not a direct FIR baseline.

## III. Problem Formulation

### A. Exact-Quota Width Assignment

Let $\mathbf{h} = [h_1, \ldots, h_N]$ be a fixed real-valued FIR coefficient vector and let $\mathcal{W} = \{16, 8, 4\}$ be the set of allowed binary widths. Given integer quotas $n_{16}$, $n_8$, $n_4$ with $n_{16}+n_8+n_4 = N$, the problem is to find the width assignment $\mathbf{w} \in \mathcal{W}^N$ minimizing (1) subject to

$$|\{i : w_i = w\}| = n_w, \quad w \in \mathcal{W}. \tag{2}$$

The quantization operator $Q(h, w)$ rounds $h$ to the nearest value representable with $w$ bits (1 sign, $w-1$ fractional bits, no saturation within the tested range).

### B. Equivalence to Unit-Capacity Transportation

Problem (1)–(2) is a binary assignment MILP. Let $x_{i,k} \in \{0, 1\}$ indicate that coefficient $i$ receives width choices$(k)$; the cost of assignment $x_{i,k} = 1$ is $c_{i,k} = (h_i - Q(h_i, \text{choices}(k)))^2$. Then:

$$\min_{\mathbf{x}} \quad \sum_{i=1}^{N} \sum_{k=1}^{3} c_{i,k} x_{i,k} \tag{3}$$

$$\text{s.t.} \quad \sum_{k} x_{i,k} = 1, \quad i = 1, \ldots, N \tag{4}$$

$$\sum_{i} x_{i,k} = n_k, \quad k = 1, 2, 3 \tag{5}$$

$$x_{i,k} \in \{0, 1\} \tag{6}$$

is a unit-capacity bipartite transportation / min-cost flow problem. The network has a source connected to each of the $N$ coefficient nodes (capacity 1, cost 0), each coefficient node connected to each of the 3 width nodes (capacity 1, cost $c_{i,k}$), and each width node connected to a sink (capacity $n_k$, cost 0). Sending $N$ units of flow minimizes total cost and satisfies all quotas exactly.

### C. Reduction to Scalar Exact Integer Cost

Llorens et al. [4] study the minimization of an L-alpha coefficient-vector error under a scalar exact integer total cost $\sum_i r_{w_i} = R^*$, solved by a constrained-shortest-path DP. The present three-width exact-quota problem is a special case. Set $B = N + 1$, $r_{16} = B + 1$, $r_8 = 2$, $r_4 = 1$. Then for any assignment, $R = N + Bn_{16} + n_8$; because $0 \le n_8 < B$, the value $R^* = N + Bn_{16} + n_8$ uniquely encodes the quotas $(n_{16}, n_8, n_4)$. The Llorens DP at exact cost $R^*$ therefore solves the quota-constrained problem. This was empirically verified on all 108 saved test configurations (maximum objective discrepancy: floating-point rounding).

The DP developed in Section IV is a direct state encoding that tracks used quota counts rather than scalar cost; it is equivalent in result but distinct in state representation. Neither formulation constitutes a new optimization class.

## IV. DP Algorithm and Greedy Baseline

### A. Dynamic Programming

The DP processes coefficients in order and stores the minimum partial cost for each reached quota state. Let $D[i][c_{16}][c_8][c_4]$ denote the minimum squared error for the first $i$ coefficients using exactly $c_{16}$ 16-bit, $c_8$ 8-bit, and $c_4$ 4-bit assignments. The recurrence is:

$$\begin{aligned} D[i][c_{16}][c_8][c_4] = \min\big\{ & D[i-1][c_{16}-1][c_8][c_4] + e_i(16), \\ & D[i-1][c_{16}][c_8-1][c_4] + e_i(8), \\ & D[i-1][c_{16}][c_8][c_4-1] + e_i(4) \big\} \end{aligned} \tag{7}$$

where $e_i(w) = (h_i - Q(h_i, w))^2$, with the constraint that each dimension of the predecessor state is non-negative.

The time complexity is $O(N(n_{16}+1)(n_8+1)(n_4+1))$, matching the state count; each state has at most three transitions. The space complexity is the same (or can be reduced to one layer by processing coefficients in reverse order).

**Optimality.** By induction on $i$: assume $D[i-1]$ stores the optimal cost for each reachable quota state. At step $i$, the recurrence examines every feasible predecessor; because the objective is separable ($\sum_i e_i(w_i)$) and each coefficient chooses exactly one width, the extension from the optimal predecessor yields the optimal value at $D[i]$. The terminal state $D[N][n_{16}][n_8][n_4]$ is therefore the global minimum for the stated objective and quota constraints.

### B. Greedy Baseline

The greedy comparator starts with every coefficient at 4 bits and repeatedly selects the available 4-to-8 or 8-to-16 upgrade with the greatest immediate reduction in squared coefficient error, subject to remaining quota. It enforces the same final quota counts as the DP.

The greedy algorithm was implemented as `adaptive_precision_error_greedy.m`. It is described in Section V alongside the 108-configuration test results.

## V. Software Validation

### A. Exhaustive Consistency Checks

The DP implementation was checked against exhaustive enumeration on 29 selected small cases (3–5 taps, various quotas). All costs agreed within floating-point tolerance (maximum absolute discrepancy $\sim 10^{-15}$). The greedy implementation also matched the exhaustive optima on those 29 cases.

### B. 108-Configuration Test Matrix

A larger matrix covers 3 filter types (lowpass/highpass/bandpass) $\times$ 4 cutoff/center settings $\times$ 3 tap counts (8/16/32, orders 7/15/31) $\times$ 3 quotas, yielding 108 configurations. These are configurations of the same underlying approach, not 108 independent filter designs.

Across the 108 cases:

- DP and greedy both satisfy quota constraints in every case.
- The greedy matched the DP objective in 85/108 = 78.7% of configurations; median excess cost 0%, mean 37.77%, max 2166.37%.
- Maximum absolute objective gap: $2.012 \times 10^{-6}$.

These are descriptive statistics; they do not imply a general performance claim for the greedy method.

Timing was measured on MATLAB R2025a (Intel Core i5-12500H) with warmup and five repetitions. Median DP runtime was 0.047 ms; median greedy was 0.183 ms. A separate paired assignment-baseline run measured median DP 0.02615 ms and MILP 2.3322 ms. These are implementation-specific timings; they do not establish general runtime advantages.

TABLE I
Eight-tap routed FPGA resources (corrected RTL)

| Run | DSP | $\text{LUT}_{\text{logic}}$ | $\text{LUT}_{\text{mem}}$ | FF | Core WNS (ns) |
|---|---|---|---|---|---|
| A1/A2 | 8 | 100 | 0 | 109 | 5.952 |
| B1 | 6 | 124 | 0 | 189 | 6.500 |
| B2 | 4 | 166 | 16 | 279 | 6.725 |
| C1/C2 | 6 | 132 | 0 | 175 | 6.323 |

### C. Exact Assignment Baseline

All 108 configurations were solved as binary assignment MILP using MATLAB `intlinprog` (R2025a Optimization Toolbox). Every solve returned a positive exit flag, zero absolute and relative MIP gap, and used one branch-and-bound node. The maximum absolute objective discrepancy between MILP and DP across all 108 cases was $4.34 \times 10^{-19}$ (floating-point evaluation). This confirms the equivalence of the DP and the exact assignment formulation but does not evidence algorithmic novelty.

## VI. FPGA Case Studies

### A. Eight-Tap Filter

An 8-tap direct-form FIR was implemented in Verilog for three coefficient sets: A (all 16-bit), B (all 8-bit), and C ($[4, 16, 8, 8, 8, 8, 16, 4]$). Bit-exact integer-reference simulation covered 157 cycles with zero mismatches for all three configurations.

Six synthesis/implementation runs (A1/A2, B1/B2, C1/C2; forced-DSP and auto-mapping) were completed on XC7A35T-2FGG484 with Vivado 2022.2.2 using a 10 ns clock and assumed 2 ns I/O delays. Table I reports routed resources.

Core register-to-register WNS is positive in all runs; overall WNS is negative ($-0.213$ to $-0.695$ ns) on paths ending at output ports under assumed 2 ns output delay. These are not board-level timing signoff results. FIR-only DRC reports NSTD-1 and UCIO-1 Critical Warnings on unconstrained top-level ports.

**Board verification.** Three bitstream configurations (A, B, C) were programmed to a legacy ACX750 development board (XC7A35T-2FGG484) via JTAG. UART-paced testing delivered the same 157-sample input sequence to each configuration; each was freshly programmed and tested twice. Across 3 configurations $\times$ 2 runs $\times$ 157 samples = 942 bit-exact

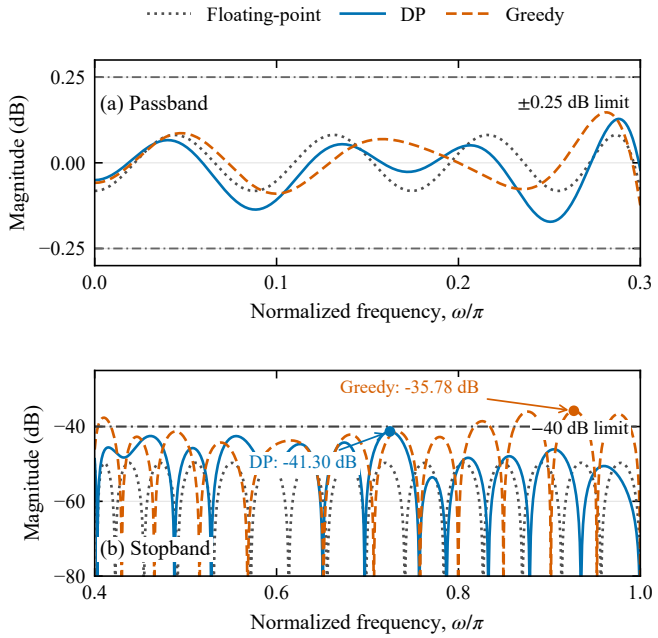



Fig. 1. Computed frequency responses of the 48-tap FIR for quota $[24, 16, 8]$: (a) passband detail and (b) stopband detail. Gray dash-dot lines mark the stated limits. DP and greedy stopband peaks are $-41.30$ and $-35.78$ dB, respectively. These curves are calculated from the saved coefficients, not measured on hardware.

comparisons against integer-precision reference values, zero mismatches were observed. This establishes functional correctness under the tested operating point; it does not measure maximum frequency, throughput, or power.

### B. Forty-Eight-Tap Filter

A 48-tap equiripple low-pass FIR (passband 0–0.3$\pi$, stopband 0.4$\pi$–$\pi$, $\pm 0.25$ dB passband, $\leq -40$ dB stopband) was designed with MATLAB `firpm`. For quota $[24, 16, 8]$, the DP allocation has coefficient SSE $1.8594 \times 10^{-5}$ and a computed stopband peak of $-41.30$ dB (meets the $-40$ dB limit); a same-quota greedy allocation has SSE $6.6658 \times 10^{-5}$ and $-35.78$ dB (fails). Two other quotas ($[16, 24, 8]$ and $[12, 24, 12]$) fail the stopband limit even under DP, showing that coefficient-SSE optimality does not guarantee spectral compliance.

Figure 1 compares the computed responses for the first quota. Both allocations meet the passband limit, but only DP meets the stopband limit in this case.

Eight RTL variants (DP, greedy, uniform 16-bit, uniform 8-bit; direct and symmetric-folded) passed XSim simulation over 2,500 cycles with 1,861 valid outputs per design (14,888 total output comparisons) against an independent 48-term integer-convolution reference. Four symmetric-folded and direct-form routes were completed on the same XC7A35T part. Table II reports routed resources for the DP and greedy symmetric-folded designs.

The response-compliant DP allocation uses more slice LUTs (354 vs. 216) and flip-flops (1066 vs. 997) than the non-compliant greedy allocation; this is a bounded resource–response tradeoff for one filter and one quota, not a general

TABLE II
FORTY-EIGHT-TAP ROUTED RESOURCES FOR QUOTA $[24, 16, 8]$

| Design | DSP | $\text{LUT}_L$ | $\text{LUT}_M$ | Slice LUT | FF | Reg–reg (ns) |
|---|---|---|---|---|---|---|
| DP, folded | 21 | 322 | 32 | 354 | 1066 | 4.344 |
| Greedy, folded | 21 | 200 | 16 | 216 | 997 | 5.189 |
| DP, direct | 38 | 654 | 80 | 734 | 1106 | 5.763 |
| Greedy, direct | 39 | 412 | 64 | 476 | 1043 | 4.879 |

Overall WNS (row order): $-0.213$, $-0.430$, $-0.435$, $-0.213$ ns. Negative WNS on assumed output-delay paths.

TABLE III
ROBUSTNESS STUDY: COEFFICIENT SSE AND COMPUTED STOPBAND PEAK

| Case | Quota | DP SSE | Greedy SSE | DP stop (dB) | DP pass? | Greedy pass? |
|---|---|---|---|---|---|---|
| A | $[24, 16, 8]$ | $1.86 \cdot 10^{-5}$ | $6.67 \cdot 10^{-5}$ | $-41.30$ | Yes | No |
| A | $[16, 24, 8]$ | $5.61 \cdot 10^{-5}$ | $1.07 \cdot 10^{-4}$ | $-35.32$ | No | No |
| A | $[12, 24, 12]$ | $9.93 \cdot 10^{-5}$ | $1.25 \cdot 10^{-4}$ | $-31.45$ | No | No |
| B | $[24, 16, 8]$ | $4.02 \cdot 10^{-5}$ | $8.42 \cdot 10^{-5}$ | $-35.55$ | No | No |
| B | $[16, 24, 8]$ | $9.35 \cdot 10^{-5}$ | $1.24 \cdot 10^{-4}$ | $-34.02$ | No | No |
| B | $[12, 24, 12]$ | $1.56 \cdot 10^{-4}$ | $1.56 \cdot 10^{-4}$ | $-32.42$ | No | No |
| C | $[24, 16, 8]$ | $5.47 \cdot 10^{-5}$ | $9.62 \cdot 10^{-5}$ | $-34.86$ | No | No |
| C | $[16, 24, 8]$ | $8.18 \cdot 10^{-5}$ | $1.16 \cdot 10^{-4}$ | $-35.32$ | No | No |
| C | $[12, 24, 12]$ | $1.32 \cdot 10^{-4}$ | $1.51 \cdot 10^{-4}$ | $-31.85$ | No | No |
| D | $[18, 12, 6]$ | $2.22 \cdot 10^{-5}$ | $2.22 \cdot 10^{-5}$ | $-35.60$ | No | No |
| D | $[12, 18, 6]$ | $5.94 \cdot 10^{-5}$ | $5.94 \cdot 10^{-5}$ | $-30.75$ | No | No |
| D | $[10, 16, 10]$ | $3.02 \cdot 10^{-4}$ | $3.02 \cdot 10^{-4}$ | $-28.16$ | No | No |
| E | $[32, 22, 10]$ | $2.20 \cdot 10^{-5}$ | $7.67 \cdot 10^{-5}$ | $-40.76$ | Yes | No |
| E | $[22, 32, 10]$ | $5.50 \cdot 10^{-5}$ | $8.08 \cdot 10^{-5}$ | $-36.78$ | No | No |
| E | $[16, 32, 16]$ | $8.68 \cdot 10^{-5}$ | $9.68 \cdot 10^{-5}$ | $-32.85$ | No | No |

superiority claim. No 48-tap board test, measured Fmax, sustained throughput, or power measurement was performed.

## VII. Robustness Study: Five Filter Specifications

### A. Designs and Quotas

To assess whether the 48-tap base-case observation generalizes, five FIR designs were tested under three proportional quotas each (15 design/quota combinations). The designs are:

- A: 48-tap, passband 0–0.3$\pi$, stopband 0.4$\pi$–$\pi$, $-40$ dB (base case)
- B: 48-tap, same bands, $-50$ dB stopband limit
- C: 48-tap, passband 0–0.2$\pi$, stopband 0.3$\pi$–$\pi$, $-40$ dB (narrower transition)
- D: 36-tap, same bands and limit as A (replaced the originally planned 32-tap case, which did not meet $-40$ dB as a floating-point prototype)
- E: 64-tap, same bands and limit as A

All five floating-point prototypes satisfy their stated passband and stopband limits. Quotas are proportional to the 48-tap base case ($[24, 16, 8]$, $[16, 24, 8]$, $[12, 24, 12]$), scaled proportionally to the tap count and adjusted to sum exactly to that count. Some scaled quotas are odd (for example, $[22, 32, 10]$ for 64 taps).

### B. Results

Table III reports the coefficient SSE and computed stopband peak for each design/quota/method combination. DP and MILP coefficient-SSE optima agree in every case.

### C. Interpretation

DP meets both response limits in 2 of 15 combinations; greedy meets them in 0. The two DP passes occur for

the highest-precision quota in cases A and E. This is not a universal frequency-response advantage: in case C's first quota, greedy has a slightly lower stopband peak ($-35.37$ vs. $-34.86$ dB), although both fail the $-40$ dB limit.

Lower coefficient SSE does not imply lower stopband peak or compliance with a response specification. These results support a bounded observation about five prototypes and three proportional quotas only; they do not establish general performance claims. All results are software frequency-response calculations; the 36- and 64-tap designs were not synthesized, routed, or tested on hardware.

## VIII. Response-Aware Allocation: Feasibility Investigation

This section documents an investigation into whether the coefficient-SSE objective could be replaced by a frequency-response-aware objective, and whether the resulting allocation could be solved exactly under exact quota constraints.

### A. Motivation and Mathematical Structure

The coefficient-SSE objective $\sum_i e_i^2$ is (by Parseval's theorem) proportional to the unweighted integrated squared frequency-response error over the full band. To focus on stopband compliance, the natural objective is the stopband-weighted integrated squared complex-response error $\mathbf{e}^\top R\mathbf{e}$, where

$$R_{ij} = \frac{1}{\pi}\int_{\Omega_{\text{stop}}} \cos(\omega(i-j))\, d\omega \tag{8}$$

and $\Omega_{\text{stop}}$ is the stopband interval.

Three key properties were established:

1) $R_{ii}$ is the same for all tap positions $i$: the diagonal weight equals the stopband bandwidth fraction and does not vary with tap index. Diagonal approximation therefore produces a uniform scaling of the SSE objective and does not change the allocation.
2) $R$ has nonzero off-diagonal elements (magnitudes $6 \times 10^{-6}$ to $0.30$ for the 48-tap case), so the objective $\mathbf{e}^\top R\mathbf{e}$ is not separable in the tap errors $e_i$. The existing quota DP, which assumes a separable objective, cannot directly solve the weighted formulation.
3) Fang et al. [11] also studied FIR allocation with a frequency-domain objective (minimax), so the objective type alone does not establish novelty.

### B. Small-Scale Exhaustive Screening

To assess whether handling the cross-terms changes the optimal allocation, exhaustive search was run on five 8-tap filters under two quotas each (10 cases). Each case enumerated all feasible assignments (420–560 per case), found the SSE-minimizing allocation and the $\mathbf{e}^\top R\mathbf{e}$-minimizing allocation, and compared them.

In 6 of 10 cases, the two optima differed and the weighted-optimal allocation had a lower stopband peak. However, the absolute stopband peaks for these 8-tap filters are poor ($-5$ to $-20$ dB), and the improvements are small (0.2–1.2 dB). The 10 cases arise from 5 filter designs $\times$ 2 quotas; independence is limited.

### C. Forty-Eight-Tap MIQP Attempt

To test whether the weighted objective can be solved exactly at realistic filter scales, the objective was linearized using binary product variables. For each pair $(i,k),(j,l)$ with $i<j$, a product variable $y = x_{i,k}x_{j,l}$ was introduced with standard McCormick inequalities. The resulting MILP had:

- 144 width-selection binary variables
- 10,152 product binary variables
- 51 equality constraints
- 30,456 McCormick inequality constraints

MATLAB `intlinprog` was warm-started with the SSE-DP solution. After approximately 93 seconds, the root-node LP lower bound was approximately $-5.637 \times 10^{-4}$ against an incumbent of $7.4603 \times 10^{-6}$ (gap $\approx 7{,}656\%$), with no branch-and-bound nodes explored. The run was manually interrupted after approximately 93 seconds under the pre-declared slow-progress stopping condition. The root bound changed negligibly from approximately $-5.6371 \times 10^{-4}$. Ctrl+C terminated MATLAB before it returned an exit flag or final allocation. The negative LP bound reflects weak LP relaxation from the McCormick linearization; it is not a property of the objective.

### D. Summary and Decision

Under the pre-declared stopping conditions (stop if progress is too slow; otherwise cap the run at 1 hour), this branch was closed:

> *STOP: The direct pairwise-product MILP has a weak LP relaxation that did not certify global optimality in the observed run. No response-aware allocation or improvement claim follows from the attempted runs.*

This outcome does not imply that response-aware allocation with exact quota constraints is mathematically intractable. A dedicated MIQP solver (e.g., Gurobi or CPLEX) or a stronger relaxation strategy may certify global optimality at 48-tap scale. This remains future work conditioned on solver availability.

## IX. Conclusion

This report documented the design, analysis, and experimental validation of exact-quota mixed-precision FIR coefficient quantization.

The central problem—assign $N$ coefficients to three fixed-point widths under exact count constraints, minimizing squared coefficient error—is a unit-capacity bipartite transportation problem. Its DP solution is a direct state-encoding realization of known cost-constrained FIR quantization DP; it is not a new optimization class or principle. This limitation was established mathematically and confirmed empirically by scalar-cost encoding and full MILP comparison on all 108 test cases.

The implemented DP achieves the global minimum of the stated objective, as verified by 29 exhaustive checks and 108-case MILP comparison (maximum objective discrepancy: $4.34 \times 10^{-19}$). Three levels of hardware validation were

completed for an 8-tap implementation: 6 FPGA synthesis/implementation runs on Artix-7, and 942 bit-exact board functional comparisons. A 48-tap case study added software allocation comparison, RTL simulation (14,888 output comparisons), and 4 FPGA synthesis runs; in the 48-tap case, DP allocation meets a $-40\,\mathrm{dB}$ stopband limit under one quota where same-quota greedy does not, at the cost of more LUTs and registers.

A robustness study across 5 filter designs and 15 quotas showed DP meeting the stated stopband limit in 2 of 15 cases, greedy in 0. DP does not universally produce lower stopband peaks than greedy; in some cases greedy's peak is slightly lower, although both fail the limit. Coefficient-SSE optimality does not imply spectral compliance.

A response-aware allocation branch was explored: the band-weighted objective is not separable (cross-terms exist), the diagonal approximation is uniform and does not change the allocation, and the 48-tap MIQP under MILP McCormick linearization showed a weak LP relaxation and was manually stopped after about 93 seconds without proving optimality. MATLAB returned no exit flag or allocation. This branch was closed under the pre-declared slow-progress stopping condition.

No novelty claim is made. The work demonstrates a complete chain of reproducible evidence—software, FPGA, and hardware—for one 8-tap and one 48-tap filter case. It includes both the technical findings and the negative feasibility results from the response-aware allocation investigation.

**Future work.** Response-aware allocation with exact quotas may be tractable with dedicated MIQP solvers (Gurobi, CPLEX) or tighter reformulations; the present toolchain did not include those solvers. Physical board testing, Fmax characterization, and power measurement for the 48-tap design remain open. A wider filter and quota study with hardware validation would strengthen the characterization of algorithm-selection tradeoffs in practical designs.

## Code and Data Availability

The source code, experimental data, FPGA RTL, and test vectors are available at https://github.com/Bairuirui68/exact-quota-mixed-precision-fir.

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